

Chapter 20 Crustaceans

Zoology

Instructor

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Outline

Subphylum Crustacea (甲壳亚门)

Form and Function (Crayfish Model)

Classification of Crustaceans

Class Malacostraca (软甲纲)

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General Characteristics of Crustaceans

- “Insects of the Sea” (海洋中的昆虫).
- Appendages (附肢):
 - Two pairs of antennae (两对触角) (Unique to crustaceans).
 - Mandibles (大颚) and two pairs of maxillae (小颚).
 - Ancestrally biramous (双肢型).
- Respiration (呼吸): Gills (鳃) or body surface.
- Tagmata (体部): Head (头), Thorax (胸), Abdomen (腹) (or Cephalothorax + Abdomen).
- Carapace (背甲): Dorsal cuticle often covers cephalothorax.

Crustacean Phylogeny

- Cladogram showing relationships of Crustacea and Hexapoda (Pancrustacea).
- Crustaceans are divided into several classes (e.g., Malacostraca, Branchiopoda, Maxillopoda).



Figure 1: Cladogram showing hypothetical ancestor-descendant relationships of hexapods and classes of crustaceans. Hexapods and crustaceans are hypothetical sister groups evolving from a common ancestor defined by numerous shared derived characteristics. Characters followed by a question mark may be ancestral rather than shared derived features. The acron is the anterior region of the head and is not counted as a segment.

Class Malacostraca Body Plan (软甲纲体型)

- Largest class (lobsters, crabs, shrimp).
- Head: 5 fused segments.
- Thorax: 8 segments.
- Abdomen: 6 segments.
- Telson (尾节): Posterior end.
- Rostrum (额角): Anterior projection.



Figure 2: Archetypical plan of Malacostraca. The two maxillae and three maxillipeds have been separated diagrammatically to illustrate the general plan.

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External Features (Crayfish)

- Cuticle (角质层): Chitin, protein, and calcium carbonate (calcified).
- Tergum (背板) and Sternum (腹板).
- Cephalothorax (头胸部) covered by carapace.
- Abdomen (腹部) ends in telson and uropods (tail fan).



Figure 3: External structure of crayfishes. A, Dorsal view. B, Ventral view.

Appendages and Serial Homology

- Serial Homology (连续同源):
Appendages evolved from a common biramous form.
- Biramous Structure:
 - Protopod (原肢) (Coxa + Basis).
 - Exopod (外肢) (Lateral).
 - Endopod (内肢) (Medial).



Figure 4: Parts of a biramous crustacean appendage (third maxilliped of a crayfish). The two branches of the appendage are the exopod and the endopod; both extend from the protopod.

Appendage Variations

- Modifications for different functions (sensing, feeding, walking, swimming).
- Note the reduction or loss of exopods in walking legs.



Figure 5: Appendages of a crayfish showing variations on the basic biramous plan, as found in a swimmeret. Protopod, brown; endopod, blue; exopod, yellow.

Summary of Crayfish Appendages

- Table showing the diversity of function from the basic plan.
- Antennae, Mandibles, Maxillae, Maxillipeds, Chelipeds, Walking legs, Swimmerets, Uropods.



Figure 6: Crayfish Appendages. Shows how various appendages have become modified from the presumed ancestral biramous plan and now perform disparate functions.

Internal Anatomy (Crayfish)

- Muscular System: Striated muscles (Flexors/Extensors).
- Digestive System: Stomach with Gastric Mill (胃磨).
- Nervous System: Supraesophageal ganglia (Brain), ventral nerve cord.



Figure 7: Internal structure of a male crayfish.

Circulatory and Respiratory Systems

- Open Circulatory System (开放式循环).
- Hemocoel (血腔).
- Heart with ostia (心孔).
- Gills (鳃): Featherlike, attached to legs/body wall.
- “Bailer” (scaphognathite) of 2nd maxilla pumps water.



Figure 8: Diagrammatical cross section through heart region of a crayfish showing direction of blood flow in this “open” blood system. Heart pumps blood to body tissues through arteries, which empty into tissue sinuses. Returning blood enters sternal sinus, then goes through gills for gas exchange, and finally back to pericardial sinus by efferent channels. Note absence of veins.

Excretory System

- Antennal Glands (Green Glands) (触角腺/绿腺).
- Located at base of antennae.
- Function: Osmoregulation (fluid control) and ionic regulation.
- Nitrogenous waste (ammonia) diffuses across gills.



Figure 9: Scheme of antennal gland (green gland) of crayfishes. (In natural position organ is much folded.) Some crustaceans lack a labyrinth, and the excretory tubule (nephridial canal) is a much-coiled tube.

Sensory Systems: Compound Eye

- Composed of Ommatidia (小眼).
- Apposition Image (Mosaic) (并列像): Day vision; pigment isolates ommatidia.
- Superposition Image (Continuous) (重叠像): Night vision; pigment retracts.



Figure 10: Portion of compound eye of an arthropod showing migration of pigment in ommatidia for day and night vision. Five ommatidia are represented in each diagram. In daytime each ommatidium is surrounded by a dark pigment collar so that each ommatidium is stimulated only by light rays that enter its own cornea (mosaic vision); in nighttime, pigment forms incomplete collars and light rays can spread to adjacent ommatidia (continuous, or superposition, image).

Reproduction and Life Cycle

- Most are dioecious (separate sexes).
- Nauplius (无节幼体): Ancestral larva (3 pairs of appendages).
- Metamorphosis (变态): Change from larva to adult.



Figure 11: Life cycle of a Gulf shrimp, *Farfantepenaeus*. Penaeids spawn at depths of 40 to 90 m. Young larval forms are planktonic and move inshore to water of lower salinity to develop as benthic juveniles and adults. Older shrimp return to deeper water offshore.

Molting (Ecdysis) (蜕皮)

- Necessary for growth.
- Premolt: Secretion of new epicuticle, dissolution of old endocuticle.
- Ecdysis: Shedding.
- Postmolt: Stretching and hardening.



Figure 12: Cuticle secretion and resorption in ecdysis.

Molting Process

- Lobster withdrawing from old exoskeleton.
- Vulnerable during this period.



Figure 13: Molting sequence in a lobster, *Homarus americanus*. A, Membrane between carapace and abdomen ruptures, and carapace begins slow elevation. This step may take up to 2 hours. B and C, Head, thorax, and finally abdomen withdraw. This process usually takes no more than 15 minutes. Immediately after ecdysis, chelipeds are desiccated and body is very soft. Lobster continues rapid absorption of water so that within 12 hours the body increases about 20% in length and 50% in weight. Tissue water will be

Feeding and Digestion

- Gastric Mill (胃磨): Calcareous teeth in stomach for grinding food.
- Filter setae strain particles.



Figure 14: Malacostracan stomach showing gastric “mill” and directions of food movements. Mill has chitinous ridges, or teeth, for mastication, and setae for straining food before it passes into the pyloric stomach.

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Class Malacostraca (软甲纲)

Classes Remipedia and Cephalocarida

- Remipedia (浆足纲): Ancestral features, cave-dwelling, homonomous trunk segments.
- Cephalocarida (头虾纲): Tiny, benthic, hermaphroditic.



Figure 15: A, A remipede crustacean of class Remipedia. B, A cephalocarid crustacean of class Cephalocarida.

Class Branchiopoda (鳃足纲)

- Phyllopodia (叶足): Flattened legs for respiration/feeding.
- Anostraca (无甲目): Fairy shrimp.
- Notostraca (背甲目): Tadpole shrimp.
- Diplostraca (双甲目): Water fleas (**Daphnia**).



Figure 16: Animals in A, B, and C are members of class Branchiopoda. A, Tadpole shrimp (order Notostraca). B, Fairy shrimp (order Anostraca). C, **Daphnia** (order Diplostraca, suborder Cladocera).

Class Ostracoda (介形纲)

- Enclosed in bivalve carapace (双瓣壳).
- Resemble tiny clams.
- Fusion of trunk segments.



Figure 17: A, An ostracod of class Ostracoda. B, A mystacocarid crustacean of subclass Mystacocarida, class Maxillopoda. C, A copepod with attached ovisacs; subclass Copepoda, class Maxillopoda.

Class Maxillopoda (颚足纲)

- Subclass Copepoda (桡足亚纲):
 - No carapace.
 - Antennules for swimming.
 - Major component of zooplankton (primary consumers).
- Subclass Mystacocarida (须虾亚纲):
Interstitial.



Figure 18: A, An ostracod of class Ostracoda. B, A mystacocarid crustacean of subclass Mystacocarida, class Maxillopoda. C, A copepod with attached ovisacs; subclass Copepoda, class Maxillopoda.

Subclass Tantulocarida (微虾亚纲)

- Tiny ectoparasites of deep-sea crustaceans.



Figure 19: A tantulocarid. This curious little parasite is shown attached to the first antenna of its copepod host at left; subclass Tantulocarida, class Maxillopoda.

Subclass Branchiura (鳃尾亚纲)

- Fish Lice (鱼虱).
- Ectoparasites on fish.
- Suction cups (modified maxillae).



Figure 20: Fish louse; subclass Branchiura, class Maxillopoda.

Subclass Pentastomida (五口亚纲)

- Tongue Worms.
- Parasites of vertebrate respiratory systems (lungs/nasal cavities).
- Wormlike, segmented appearance (annuli).
- 4 hooks + mouth.



Figure 21: Two pentastomids. A, *Linguatula*, found in nasal passages of carnivorous mammals. Female is shown with some internal structures. B, Female *Armillifer*, a pentastomid with pronounced body rings. In parts of Africa and Asia, humans are parasitized by immature stages; adults (10 cm long or more) live in lungs of snakes. Human infection may occur from eating snakes or from contaminated food or water.

Pentastomid Structure

- Anterior end showing hooks and mouth.



Figure 22: Anterior end of a pentastome. Note both the mouth (arrow), between the middle hooks, and the apical sensory papillae.

Subclass Cirripedia (蔓足亚纲/藤声)

- Barnacles.
- Sessile adults.
- Carapace secretes calcareous plates.
- Cirri (蔓足): Feathery legs for filter feeding.
- Acorn barnacles (**Balanus**) and Gooseneck barnacles (**Lepas**).



Figure 23: Barnacles; order Thoracica, subclass Cirripedia, class Maxillopoda. A, Acorn barnacles, **Balanus balanoides**, on an intertidal rock await the return of the tide. B, Common gooseneck barnacles, **Lepas anatifera**. Note the feeding legs, or cirri, on **Lepas**. Barnacles attach themselves to a variety of firm substrates, including rocks, pilings, and boat bottoms.

Parasitic Barnacles (Rhizocephala)

- **Sacculina** (根头目).
- Parasitizes crabs.
- Root-like absorptive system.
- Castrates the host.



Figure 24: Life cycle of **Sacculina** (order Rhizocephala, subclass Cirripedia, class Maxillopoda), parasite of crabs (**Carcinus**).

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Order Isopoda (等足目)

- Dorsoventrally flattened (背腹扁平).
- No carapace.
- Sessile eyes.
- Includes terrestrial Sow bugs/Pill bugs (潮虫/鼠妇).



Figure 25: A, Four pill bugs, *Armadillidium vulgare* (order Isopoda, class Malacostraca), common terrestrial forms. B, Freshwater sow bug, *Caecidotea* sp., an aquatic isopod.

Parasitic Isopods

- Some isopods are parasitic on fish.



Figure 26: An isopod parasite (*Anilocra* sp.) on a coney (*Cephalopholis fulvus*) inhabiting a Caribbean coral reef (order Isopoda, class Malacostraca).

Order Amphipoda (端足目)

- Laterally compressed (侧扁).
- No carapace.
- Sessile eyes.
- Beach hoppers, **Gammarus**.



Figure 27: Marine amphipods. A, Free-swimming amphipod, **Anisogammarus** sp. B, Skeleton shrimp, **Caprella** sp., shown on a bryozoan colony, resemble praying mantids. C, **Phronima**, a marine pelagic amphipod, takes over the tunic of a salp (subphylum Urochordata, see Chapter 23). Swimming by means of its abdominal swimmerets, which protrude from the opening of the barrel-shaped tunic, the amphipod maneuvers to catch its prey. The tunic is not seen (order Amphipoda, class Malacostraca).

Whale Parasites

- Barnacles and Cyamid amphipods (Whale lice) on a grey whale.



Figure 28: A, Head and mouth of a healthy California grey whale, *Eschrichtius robustus*, bearing its characteristic heavy load of barnacles (order Thoracica, subclass Cirripedia, class Maxillopoda) and cyamid parasites (order Amphipoda, class Malacostraca) (arrows). Note yellowish plates of baleen in mouth (p. 640). B, Cyamid parasites of grey whale. Unlike most amphipods, these are dorsoventrally flattened. They have sharp, grasping claws on their legs.

Order Euphausiacea (磷虾目)

- Krill (磷虾).
- Carapace fused with all thoracic segments but gills exposed.
- Major food source for baleen whales.



Figure 29: **Meganyctiphanes** (order Euphausiacea, class Malacostraca) “northern krill.”

Order Decapoda (十足目)

- 5 pairs of walking legs (步足).
- First pair often chelipeds (螯足) with chelae (螯).
- 3 pairs of maxillipeds.
- Crabs, lobsters, crayfish, shrimp.



Figure 30: Decapod crustaceans. A, A bright orange tropical rock crab, *Grapsus grapsus*, is a conspicuous exception to the rule that most crabs bear cryptic coloration. B, A hermit crab, *Elassochirus gilli*, which has a soft abdominal exoskeleton, lives in a snail shell that it carries about and into which it can withdraw for protection. C, A male fiddler crab, *Uca* sp., uses its enlarged cheliped to wave territorial displays and in threat and combat. D, A red night shrimp, *Rhynchocinetes rigens*, prowls caves and overhangs of coral reefs, but only at night. E, A spiny lobster, *Panulirus argus* (shown here), and the northern lobster, *Homarus americanus*, are

Decorator Crab

- Sponge crab (**Dromidia**) masking itself for camouflage.



Figure 31: Sponge crab, **Dromidia antillensis**. This crab is one of several species that deliberately mask themselves with material from their immediate environment (order Decapoda, class Malacostraca).